

The image displays a sequence of Feynman diagrams representing a mathematical identity. It begins with the sum of four terms, each featuring a vertical red line and two horizontal blue lines. The first term shows a fermion loop with external momenta p_1 and p_2 , and a wavy line with momentum k . The second term shows a similar loop with momenta $k-p_1$ and $k-p_2$. The third and fourth terms show more complex loop configurations with various momentum labels.

These four terms are summed and equated to the product of two disc integrals:

$$= \text{disc}_{p_1^2} \text{disc}_{p_2^2}$$

To the right of this equation is a diagram of a genus-2 surface (a torus with two holes). A red vertical line passes through the center, marked with a red dot. A wavy line with momentum k connects the two holes, and a fermion line with momentum p_2 is shown on the left.